4	Let *H* be to initial height and *h* be the height at time *t.* $v^2 = u^2 + 2as$ $v^2 = 2g(H-h)$ $v^2 + 2gh = 2gH$ $h = -\frac{1}{2g}v^2 + 2gH$ $y = -k_1 x^2 + k_2$ Hence it is a negative quadratic graph with vertical intercept k_2
5	When 3 non-parallel forces act on body, all 3 must intersect at a point to achieve rotational equilibrium